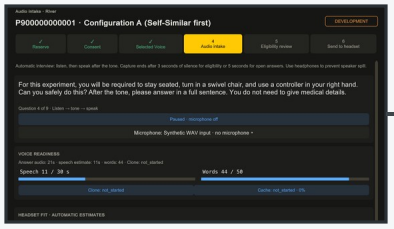
Before the session



Audio Intake
(Nine spoken prompts, Trial Review)

Clone and Synthesize
(ElevenLabs)

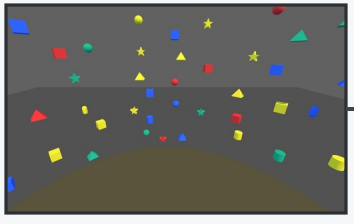


Selected Agent Voice

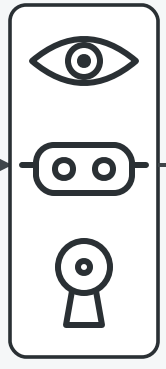
Self-Similar Voice

Prepared Phrase Libraries
(Sent to the headset ahead of time)

During each trial



Search Scene
(Unity on the HTC VIVE Focus Vision)



Gaze, Head, Controller
(Headset sensors)

Guidance State

- Gaze valid and recent
- Angle to target, trend
- Direction to target
- Recent gaze direction
- Last cue and response
- Past cues and trials
- Allowed strategies

Summarize, Then Ask
(Unity on the headset)

Compact state, not raw gaze

Wireless ADB link

Two Decisions

1. Strategy
Choose one allowed: stay silent, orient, refine, correct, or encourage

2. Phrase
Rank the prepared phrases using their recent use

Decision Model
(Local MiniLM on the lab computer)

On our computer, no cloud

Chosen phrase

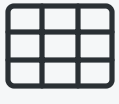


Spoken Cue
(Assigned voice)

What actually played feeds back into memory

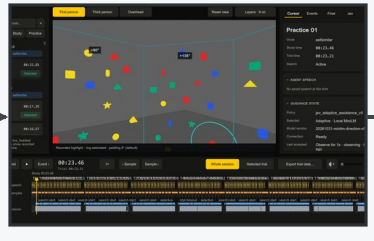
Logged every frame
Starting angle θ_0 saved each trial

After the session



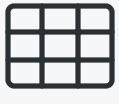
Session Recordings
(Gaze, events, replay)

Transfer



Trial Review
(Replay and live monitor)

Export



Trial CSVs